

ABDULLAH KHAWAJA

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[LinkedIn](#) | [GitHub](#) | [Hugging Face](#) | [Substack](#) | [HackerNoon](#)

TECHNICAL ARSENAL

GenAI & LLMs:	Unsloth, LangChain, Ollama, Claude Code, OpenAI, Cursor, Gemini-CLI, Fine-tuning (Gemma/Llama).
Machine Learning:	TensorFlow, Keras, PyTorch, Scikit-Learn, XGBoost, Vertex AI, BigQuery, Recommender Systems (Wide & Deep, LTR).
Development:	Python, Java, C++, React, TypeScript, Electron, SQL, Docker, Git.
Cloud & Tools:	Alibaba Cloud, AWS, GCP, Vercel, Postman, Hetzner, Tableau.

WORK EXPERIENCE

Machine Learning Engineer | DPD Group

Nov 2023 - Present

- **Consumer Chatbot Modernization:** Fine-tuned **Gemma 4 E4B IT** on **20K+ internal customer-chatbot conversations**, including tool calls and custom databooks recalled at response time, replacing Gemini-based serving with economical, Google-native inference on Vertex AI; business forecasts project **65% lower chatbot spend**.
- **Customer Retention (Churn Prediction):** Architected a 6-model meta-ensemble (including Dual LightGBMs, WaveNet, Logistic Hazard, and a Hybrid XGBoost-Transformer) that achieved the **highest predictive performance across DPD Group**. Accurately identified **64% of flagged customers** as true churners and captured **67% of total churn cases**, with **savings of \$650K recorded so far**.
- **Volume Forecasting:** Developed a deep learning model (Mixture of Experts, RNN, WaveNet) surpassing a 12-year human benchmark by **+8%**. Replaced fixed lags with **event-based lags** (Amazon Prime Day, Bank Holidays) to capture seasonality accurately, delivering consistent daily accuracy for 80+ hubs.

Data Scientist (Contract) | Dastgyr (Remote)

Jun 2023 - Nov 2023

- **Recommender Systems:** Implemented SWING collaborative filtering from scratch for the "Top For You" module, driving a **+1.44% uplift in basket size** and **+22% increase in average order value**.
- **Strategy:** Defined the AI roadmap (sales forecasting, OOS reduction) and fostered a culture of experimentation and documentation.

Senior Data Scientist - Recommendations | Daraz (Alibaba Group)

Oct 2020 - Feb 2023

- **Rank Stage:** Replaced logistic regression with **Wide & Deep** learning models, achieving **+3% CTR uplift** via advanced feature engineering and DL infrastructure integration.
- **Model Optimization:** Reduced feature space by 50% using feature importance, slashing backup recommendation compute time from **3h to 1h 25m** with no metric degradation (handling 60% of all recommendations).
- **Learning to Rank:** Introduced pointwise LTR to handle discount price/ratio trade-offs, driving a **+35% increase in conversion rate (CR)**.
- **Leadership:** Led 50+ A/B tests across 5 countries; top-rated performer for 2 consecutive years.

PROJECTS



Sort Moments | [GitHub](#)

- **Tech:** Python, PyQt6, RetinaFace, Buffalo Large (Embeddings).
- **Description:** A privacy-first desktop application engineered to tackle the chaos of large-scale event photography (4,000+ images). Public site shows **407 downloads** and **2,501 visitors**. It runs a fully local ML pipeline (RetinaFace + InsightFace) using DirectML to detect and cluster faces, automatically organizing solo and group shots by person without data ever leaving the device.
- **Features:** Offline processing, rapid image size reduction, high-precision face clustering, and a streamlined workflow for manual refinement and deduplication.



WeaveSkip

- **Tech:** Next.js, TypeScript, Prisma, PostgreSQL, Redis/BullMQ, NextAuth, OAuth, S3, Stripe, Zod.
- **Description:** A creator distribution OS that turns one article, RSS feed, or manual source into platform-native drafts for LinkedIn, X, Reddit, Facebook, Threads, and Substack Notes. Public site shows **5,344 visitors**, **3 signups**, **8 post batches**, **100% kept without edits**, and 2 finished posts.
- **Features:** Source ingestion, intent/keyword/subreddit intelligence, BYOK LLM config with encrypted keys, human review, guardrails, queue-based scheduling/publishing, and Medium crosspost support.



NeighbourFit

- **Tech:** React, TypeScript, FastAPI, PostgreSQL/PostGIS, Mapbox, OpenAI, scikit-learn.
- **Description:** A UK rental postcode checker and area-fit tool. Public counter shows **136 postcode lookups**; Map Intelligence and UK postcode checks are open while profile, saved searches, and agentic rental discovery stay in private beta.
- **Features:** Commute and amenity context, geospatial scoring, maps, source-aware caveats, saved-list flow, and fit breakdowns that show context rather than pretending to give a verdict.



Hugging Face Model Releases

- **Tech:** Gemma, Qwen, LoRA/fine-tuning, ablation, distillation, GGUF quantization.
- **Description:** Public model releases exploring small-model adaptation, ablation experiments, Opus-distilled outputs, and deployment-ready GGUF packaging.
- **Focus:** Hands-on LLM training, model surgery, quantization, and open-source distribution across multiple experimental releases.



Navigating Noise | [HackerNoon](#)

- **Focus:** Technical writing on ML systems, AI tools, benchmarks, interviews, and product lessons for engineers and builders.
- **Description:** Converts noisy AI developments into practical mental models: what to follow, how to prepare, and how to judge tools that claim to change workflows.

EDUCATION

Master of Science in Data Science (Distinction) | University of Essex, UK

Oct 2022 - Oct 2023

Bachelor of Science in Computer Science | FAST-NU, Pakistan

Aug 2016 - Sep 2020